

Introduction to Digital Logic

Lecture 3

Gate Circuits and Boolean Equations

- Binary Logic and Gates
- Boolean Algebra
- Standard Forms

Binary Logic and Gates

- Binary variables take on one of two values.
- Logical operators operate on binary values and binary variables.
- Basic logical operators are the logic functions **AND**, **OR** and **NOT**.
- Logic gates implement logic functions.
- Boolean Algebra: a useful mathematical system for specifying and transforming logic functions.
- We study Boolean algebra as foundation for designing and analyzing **digital systems**!

Binary Variables

- Recall that the two binary values have different names:
 - True/False
 - On/Off
 - Yes/No
 - 1/0
- We use **1** and **0** to denote the two values.
- Variable identifier examples:
 - A, B, y, z, or X_1 for now
 - RESET, START_IT, or ADD1 later

Logical Operations

- The three basic logical operations are:
 - AND
 - OR
 - NOT
- AND is denoted by a dot ($\cdot$)
- OR is denoted by a plus ($+$)
- NOT is denoted by an overbar ($\bar{}$), a single quote mark ($'$) after, or ($\sim$) before the variable

Notation Examples

- **Examples:**

- $Y=A.B$ is read “Y is equal to A AND B.”
- $z=x+y$ is read “z is equal to x OR y.”
- $X=\bar{A}$ is read “X is equal to NOT A.”

- **Note: The statement:**

$1 + 1 = 2$ (read “one plus one equals two”)

is not the same as

$1 + 1 = 1$ (read “1 or 1 equals 1”).

Operator Definitions

- Operations are defined on the values "0" and "1" for each operator:

AND

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 0$$

$$1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

OR

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 1$$

NOT

$$\bar{0} = 1$$

$$\bar{1} = 0$$

Truth Tables

- *Truth table* – a tabular listing of the values of a function for all possible combinations of values on its arguments
- Example: Truth tables for the basic logic operations:

AND		
X	Y	$Z = X \cdot Y$
0	0	0
0	1	0
1	0	0
1	1	1

OR		
X	Y	$Z = X + Y$
0	0	0
0	1	1
1	0	1
1	1	1

NOT	
X	$Z = \overline{X}$
0	1
1	0

Logic Function Implementation

- Using Switches

- For inputs:

- logic 1 is switch closed
 - logic 0 is switch open

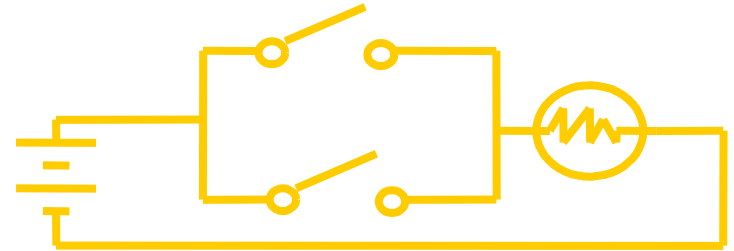
- For outputs:

- logic 1 is light on
 - logic 0 is light off.

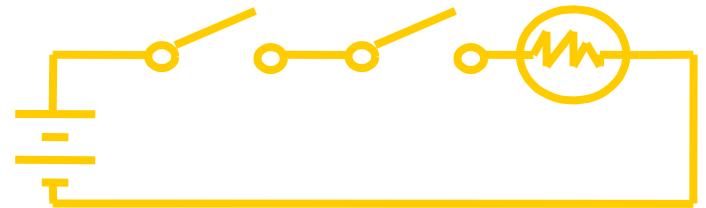
- NOT uses a switch such that:

- logic 1 is switch open
 - logic 0 is switch closed

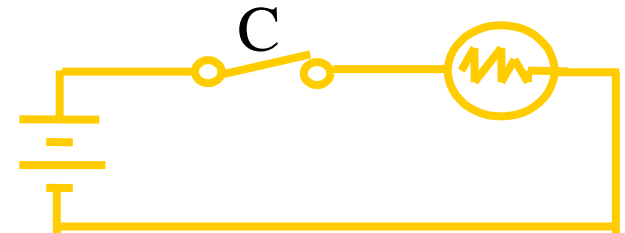
Switches in parallel => OR



Switches in series => AND

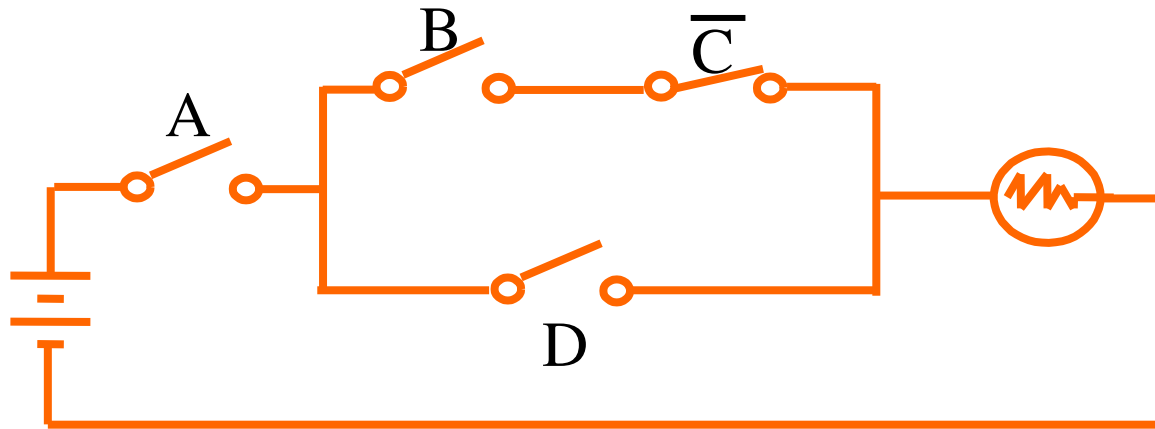


Normally-closed switch => NOT



Logic Function Implementation (Continued)

- **Example: Logic Using Switches**

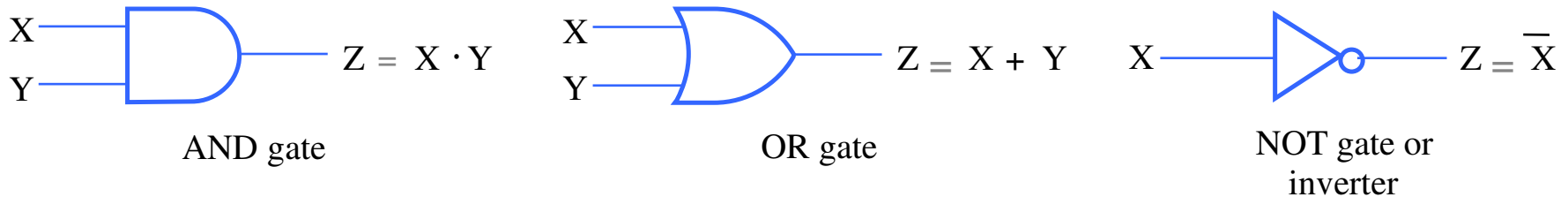


- **Light is on ($L = 1$) for**
$$L(A, B, C, D) = A \cdot ((B \cdot \overline{C}) + D)$$

and off ($L = 0$), otherwise.
- **Useful model for relay circuits and for CMOS gate circuits, the foundation of current digital logic technology**

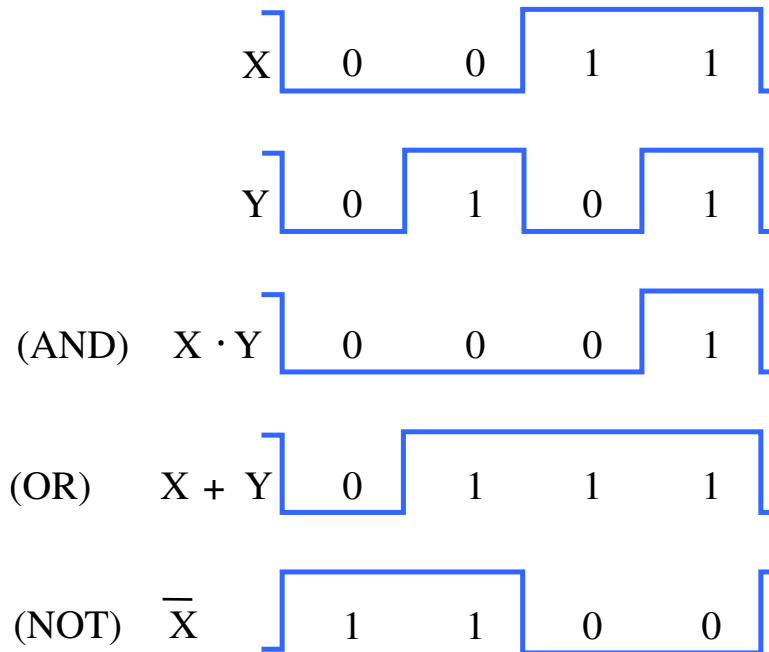
Logic Gate Symbols and Behavior

- Logic gates have special symbols:



(a) Graphic symbols

- And waveform behavior in time as follows:



(b) Timing diagram

Logic Diagrams and Expressions

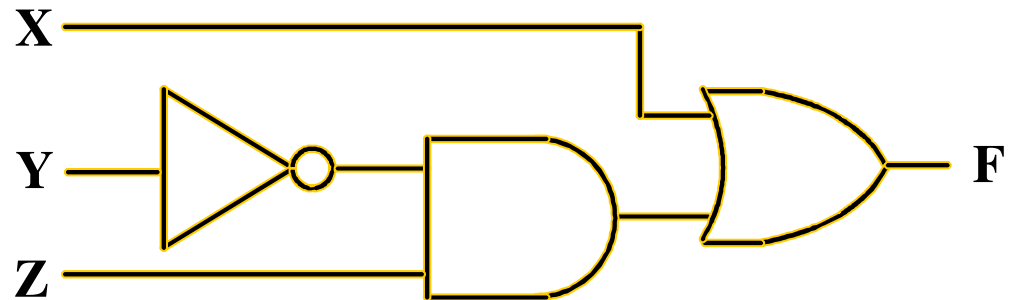
Truth Table

X Y Z	$F = X + \overline{Y} \cdot Z$
0 0 0	0
0 0 1	1
0 1 0	0
0 1 1	0
1 0 0	1
1 0 1	1
1 1 0	1
1 1 1	1

Equation

$$F = X + \overline{Y} Z$$

Logic Diagram



- Boolean equations, truth tables and logic diagrams describe the same function!
- Truth tables are unique; expressions and logic diagrams are not. This gives flexibility in implementing functions.

Boolean Algebra

- An algebraic structure defined on a set of at least two elements, B, together with three binary operators (denoted $+$, $\cdot$ and $\overline{}$) that satisfies the following basic identities:

$$1. \quad X + 0 = X$$

$$2. \quad X \cdot 1 = X$$

Existence of 0 and 1

$$3. \quad X + 1 = 1$$

$$4. \quad X \cdot 0 = 0$$

$$5. \quad X + X = X$$

$$6. \quad X \cdot X = X$$

Idempotence

$$7. \quad X + \overline{X} = 1$$

$$8. \quad X \cdot \overline{X} = 0$$

Existence of complement

$$9. \quad \overline{\overline{X}} = X$$

Involution

$$10. \quad X + Y = Y + X$$

$$11. \quad XY = YX$$

Commutative

$$12. \quad (X + Y) + Z = X + (Y + Z)$$

$$13. \quad (XY)Z = X(YZ)$$

Associative

$$14. \quad X(Y + Z) = XY + XZ$$

$$15. \quad X + YZ = (X + Y)(X + Z)$$

Distributive

$$16. \quad \overline{X + Y} = \overline{X} \cdot \overline{Y}$$

$$17. \quad \overline{X \cdot Y} = \overline{X} + \overline{Y}$$

DeMorgan's

Boolean Operator Precedence

- **The order of evaluation in a Boolean expression is:**
 1. Parentheses
 2. NOT
 3. AND
 4. OR
- **Consequence: Parentheses appear around OR expressions**
- **Example: $F = A(B + C)(C + \overline{D})$**

Example 1: Boolean Algebraic Proof

- $A + A \cdot B = A$ (Absorption Theorem)

Proof Steps

Justification (identity or theorem)

$$A + A \cdot B$$

$$= A \cdot 1 + A \cdot B$$

$$X = X \cdot 1$$

$$= A \cdot (1 + B)$$

$$X \cdot Y + X \cdot Z = X \cdot (Y + Z) \text{ (Distributive Law)}$$

$$= A \cdot 1$$

$$1 + X = 1$$

$$= A$$

$$X \cdot 1 = X$$

- Our primary reason for doing proofs is to learn:
 - Careful and efficient use of the identities and theorems of Boolean algebra, and
 - How to choose the appropriate identity or theorem to apply to make forward progress, irrespective of the application.

Example 2: Boolean Algebraic Proofs

- $AB + \bar{A}C + BC = AB + \bar{A}C$ (Consensus Theorem)

Proof Steps: **Justification (identity or theorem)**

$$\begin{aligned} & AB + \bar{A}C + BC \\ &= AB + \bar{A}C + \mathbf{1} \cdot BC \\ &= AB + \bar{A}C + (\mathbf{A} + \bar{\mathbf{A}}) \cdot BC \\ &= \mathbf{AB} + \bar{\mathbf{A}}\mathbf{C} + \mathbf{ABC} + \bar{\mathbf{A}}\mathbf{BC} \\ &= \mathbf{AB} (1+C) + \bar{\mathbf{A}}\mathbf{C} (1+B) \\ &= \mathbf{AB} \cdot 1 + \bar{\mathbf{A}}\mathbf{C} \cdot 1 \\ &= \mathbf{AB} + \bar{\mathbf{A}}\mathbf{C} \end{aligned}$$

Example 3: Boolean Algebraic Proofs

- $(\overline{X + Y})Z + X\overline{Y} = \overline{Y}(X + Z)$

Proof Steps	Justification (identity or theorem)
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$(\overline{X + Y})Z + X\overline{Y}$	
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=	
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Useful Theorems

$x \cdot y + \bar{x} \cdot y = y$	$(x + y)(\bar{x} + y) = y$	Minimization
$x + x \cdot y = x$	$x \cdot (x + y) = x$	Absorption
$x + \bar{x} \cdot y = x + y$	$x \cdot (\bar{x} + y) = x \cdot y$	Simplification
$x \cdot y + \bar{x} \cdot z + y \cdot z = x \cdot y + \bar{x} \cdot z$		Consensus
$(x + y) \cdot (\bar{x} + z) \cdot (y + z) = (x + y) \cdot (\bar{x} + z)$		
$\overline{x + y} = \bar{x} \cdot \bar{y}$	$\overline{x \cdot y} = \bar{x} + \bar{y}$	DeMorgan's Laws

Proof of Simplification

$$\mathbf{x \cdot y + \bar{x} \cdot y = y}$$

$$\mathbf{(x + y)(\bar{x} + y) = y}$$

Boolean Function Evaluation

$$F1 = xy\bar{z}$$

$$F2 = x + \bar{y}z$$

$$F3 = \bar{x}\bar{y}\bar{z} + \bar{x}y z + x\bar{y}$$

$$F4 = x\bar{y} + \bar{x}z$$

x	y	z	F1	F2	F3	F4
0	0	0	0	0	1	0
0	0	1	0	1	0	1
0	1	0	0	0	0	0
0	1	1	0	0	1	1
1	0	0	0	1	1	1
1	0	1	0	1	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	0

Expression Simplification

- An application of Boolean algebra
- Simplify to contain the smallest number of literals (complemented and uncomplemented variables):

$$\begin{aligned} & \mathbf{A B + \bar{A} C D + \bar{A} B D + \bar{A} C \bar{D} + A B C D} \\ &= \mathbf{A B + A B C D + \bar{A} C D + \bar{A} C \bar{D} + \bar{A} B D} \\ &= \mathbf{A B + A B (C D) + \bar{A} C (D + \bar{D}) + \bar{A} B D} \\ &= \mathbf{A B + \bar{A} C + \bar{A} B D = B (A + \bar{A} D) + \bar{A} C} \\ &= \mathbf{B (A + D) + \bar{A} C \quad 5 \text{ literals}} \end{aligned}$$

Complementing Functions

- Use DeMorgan's Theorem to complement a function:
 1. Interchange AND and OR operators
 2. Complement each constant value and literal
- Example: Complement $F = \bar{x}y\bar{z} + x\bar{y}\bar{z}$
 $\bar{F} = (x + \bar{y} + z)(\bar{x} + y + z)$
- Example: Complement $G = (\bar{a} + bc)\bar{d} + e$
 $\bar{G} = ?$

Overview – Canonical Forms

- **What are Canonical Forms?**
- **Minterms and Maxterms**
- **Index Representation of Minterms and Maxterms**
- **Sum-of-Minterm (SOM) Representations**
- **Product-of-Maxterm (POM) Representations**
- **Representation of Complements of Functions**
- **Conversions between Representations**

Canonical Forms

- It is useful to specify Boolean functions in a form that:
 - Allows comparison for equality.
 - Has a correspondence to the truth tables
- Canonical Forms in common usage:
 - Sum of Minterms (SOM)
 - Product of Maxterms (POM)

Minterms

- Minterms are AND terms with every variable present in either true or complemented form.
- Given that each binary variable may appear normal (e.g., x) or complemented (e.g., $\bar{x}$), there are 2^n minterms for n variables.
- Example: Two variables (X and Y) produce $2 \times 2 = 4$ combinations:

XY

(both normal)

$X\bar{Y}$

(X normal, Y complemented)

$\bar{X}Y$

(X complemented, Y normal)

$\bar{X}\bar{Y}$

(both complemented)

- Thus there are four minterms of two variables.

Maxterms

- Maxterms are OR terms with every variable in true or complemented form.
- Given that each binary variable may appear normal (e.g., x) or complemented (e.g., $\bar{x}$), there are 2^n maxterms for n variables.
- Example: Two variables (X and Y) produce $2 \times 2 = 4$ combinations:

$X+Y$

(both normal)

$X+\bar{Y}$

(X normal, Y complemented)

$\bar{X}+Y$

(X complemented, Y normal)

$\bar{X}+\bar{Y}$

(both complemented)

Maxterms and Minterms

- **Examples: Two variable minterms and maxterms.**

Index	Minterm	Maxterm
0	$\bar{x} \bar{y}$	$x + y$
1	$\bar{x} y$	$x + \bar{y}$
2	$x \bar{y}$	$\bar{x} + y$
3	$x y$	$\bar{x} + \bar{y}$

- **The index above is important for describing which variables in the terms are true and which are complemented.**

Purpose of the Index

- The index for the minterm or maxterm, expressed as a binary number, is used to determine whether the variable is shown in the true form or complemented form.
- For Minterms:
 - “1” means the variable is “Not Complemented” and
 - “0” means the variable is “Complemented”.
- For Maxterms:
 - “0” means the variable is “Not Complemented” and
 - “1” means the variable is “Complemented”.

Index Example in Three Variables

- **Example:** (for three variables)
- Assume the variables are called X, Y, and Z.
- The standard order is X, then Y, then Z.
- The Index 0 (base 10) = 000 (base 2) for three variables). All three variables are complemented for minterm 0 ($\bar{X}, \bar{Y}, \bar{Z}$) and no variables are complemented for Maxterm 0 (X,Y,Z).
 - Minterm 0, called m_0 is $\bar{X}\bar{Y}\bar{Z}$.
 - Maxterm 0, called M_0 is $(X + Y + Z)$.
 - Minterm 6 ?
 - Maxterm 6 ?

Index Examples – Four Variables

Index	Binary	Minterm	Maxterm
i	Pattern	m_i	M_i
0	0000	$\bar{a}\bar{b}\bar{c}\bar{d}$	$a + b + c + d$
1	0001	$\bar{a}\bar{b}\bar{c}d$	?
3	0011	?	$a + b + \bar{c} + \bar{d}$
5	0101	$\bar{a}b\bar{c}d$	$a + \bar{b} + c + \bar{d}$
7	0111	?	$a + \bar{b} + \bar{c} + \bar{d}$
10	1010	$a\bar{b}c\bar{d}$	$\bar{a} + b + \bar{c} + d$
13	1101	$ab\bar{c}d$	?
15	1111	$abcd$	$\bar{a} + \bar{b} + \bar{c} + \bar{d}$

Minterm and Maxterm Relationship

- Review: DeMorgan's Theorem

$$\overline{x \cdot y} = \bar{x} + \bar{y} \text{ and } \overline{\bar{x} + \bar{y}} = x \cdot y$$

- Two-variable example:

$$M_2 = \bar{x} + y \text{ and } m_2 = x \cdot \bar{y}$$

Thus M_2 is the complement of m_2 and vice-versa.

- Since DeMorgan's Theorem holds for n variables, the above holds for terms of n variables
- giving:

$$M_i = \overline{m_i} \text{ and } m_i = \overline{M_i}$$

Thus M_i is the complement of m_i .

Minterm Function Example

- **Example:** Find $F_1 = m_1 + m_4 + m_7$
- $F_1 = \bar{x} \bar{y} z + x \bar{y} \bar{z} + x y z$

x y z	index	m_1	+	m_4	+	m_7	$= F_1$
0 0 0	0	0	+	0	+	0	= 0
0 0 1	1	1	+	0	+	0	= 1
0 1 0	2	0	+	0	+	0	= 0
0 1 1	3	0	+	0	+	0	= 0
1 0 0	4	0	+	1	+	0	= 1
1 0 1	5	0	+	0	+	0	= 0
1 1 0	6	0	+	0	+	0	= 0
1 1 1	7	0	+	0	+	1	= 1

Minterm Function Example

- $F(A, B, C, D, E) = m_2 + m_9 + m_{17} + m_{23}$
- $F(A, B, C, D, E) =$

Maxterm Function Example

- **Example: Implement F1 in maxterms:**

$$F_1 = M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6$$

$$F_1 = (x + y + z) \cdot (x + \bar{y} + z) \cdot (x + \bar{y} + \bar{z}) \\ \cdot (\bar{x} + y + \bar{z}) \cdot (\bar{x} + \bar{y} + z)$$

x y z	i	$M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6 = F1$
0 0 0	0	$0 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 0$
0 0 1	1	$1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 1$
0 1 0	2	$1 \cdot 0 \cdot 1 \cdot 1 \cdot 1 = 0$
0 1 1	3	$1 \cdot 1 \cdot 0 \cdot 1 \cdot 1 = 0$
1 0 0	4	$1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 1$
1 0 1	5	$1 \cdot 1 \cdot 1 \cdot 0 \cdot 1 = 0$
1 1 0	6	$1 \cdot 1 \cdot 1 \cdot 1 \cdot 0 = 0$
1 1 1	7	$1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 1$

Maxterm Function Example

- $F(A, B, C, D) = M_3 \cdot M_8 \cdot M_{11} \cdot M_{14}$
- $F(A, B, C, D) =$

Canonical Sum of Minterms

- Any Boolean function can be expressed as a Sum of Minterms.
 - For the function table, the minterms used are the terms corresponding to the 1's
 - For expressions, expand all terms first to explicitly list all minterms. Do this by “ANDing” any term missing a variable v with a term $(v + \bar{v})$.
- Example: Implement $f = x + \bar{x} \bar{y}$ as a sum of minterms.

First expand terms: $f = x(y + \bar{y}) + \bar{x} \bar{y}$

Then distribute terms: $f = xy + x\bar{y} + \bar{x} \bar{y}$

Express as sum of minterms: $f = m_3 + m_2 + m_0$

Another SOM Example

- **Example: $F = A + \bar{B} C$**
- **There are three variables, A, B, and C which we take to be the standard order.**
- **Expanding the terms with missing variables:**
- **Collect terms (removing all but one of duplicate terms):**
- **Express as SOM:**

Shorthand SOM Form

- From the previous example, we started with:

$$F = A + \bar{B} C$$

- We ended up with:

$$F = m_1 + m_4 + m_5 + m_6 + m_7$$

- This can be denoted in the formal shorthand:

$$F(A, B, C) = \Sigma_m(1, 4, 5, 6, 7)$$

- Note that we explicitly show the standard variables in order and drop the “m” designators.

Canonical Product of Maxterms

- Any Boolean Function can be expressed as a Product of Maxterms (POM).
 - For the function table, the maxterms used are the terms corresponding to the 0's.
 - For an expression, expand all terms first to explicitly list all maxterms. Do this by first applying the second distributive law, “ORing” terms missing variable V with a term equal to $V \cdot \bar{V}$ and then applying the distributive law again.
- Example: Convert to product of maxterms:

$$f(x, y, z) = x + \bar{x} \bar{y}$$

Apply the distributive law:

$$x + \bar{x} \bar{y} = (x + \bar{x})(x + \bar{y}) = 1 \cdot (x + \bar{y}) = x + \bar{y}$$

Add missing variable z :

$$x + \bar{y} + z \cdot \bar{z} = (x + \bar{y} + z)(x + \bar{y} + \bar{z})$$

Express as POM: $f = M_2 \cdot M_3$

Another POM Example

- Convert to Product of Maxterms:

$$f(A, B, C) = A \bar{C} + B C + \bar{A} \bar{B}$$

- Use $x + y z = (x+y) \cdot (x+z)$ with $x = (A \bar{C} + B C)$, $y = \bar{A}$, and $z = \bar{B}$ to get:

$$f = (A \bar{C} + B C + \bar{A})(A \bar{C} + B C + \bar{B})$$

- Then use $x + \bar{x} y = x + y$ to get:

$$f = (\bar{C} + B C + \bar{A})(A \bar{C} + C + \bar{B})$$

and a second time to get:

$$f = (\bar{C} + B + \bar{A})(A + C + \bar{B})$$

- Rearrange to standard order,

$$f = (\bar{A} + B + \bar{C})(A + \bar{B} + C) \text{ to give } f = M_5 \cdot M_2$$

Function Complements

- The complement of a function expressed as a sum of minterms is constructed by selecting the minterms missing in the sum-of-minterms canonical forms.
- Alternatively, the complement of a function expressed by a Sum of Minterms form is simply the Product of Maxterms with the same indices.
- Example: Given $F(x, y, z) = \Sigma_m(1, 3, 5, 7)$
 $\bar{F}(x, y, z) = \Sigma_m(0, 2, 4, 6)$
 $\bar{F}(x, y, z) = \Pi_M(1, 3, 5, 7)$

Conversion Between Forms

- To convert between sum-of-minterms and product-of-maxterms form (or vice-versa) we follow these steps:
 - Find the function complement by swapping terms in the list with terms not in the list.
 - Change from products to sums, or vice versa.
- Example: Given F as before: $F(x, y, z) = \Sigma_m(1, 3, 5, 7)$
- Form the Complement: $\bar{F}(x, y, z) = \Sigma_m(0, 2, 4, 6)$
- Then use the other form with the same indices – this forms the complement again, giving the other form of the original function: $F(x, y, z) = \Pi_M(0, 2, 4, 6)$

Standard Forms

- **Standard Sum-of-Products (SOP) form:**
equations are written as an OR of AND terms
- **Standard Product-of-Sums (POS) form:**
equations are written as an AND of OR terms
- **Examples:**
 - **SOP:** $A B C + \bar{A} \bar{B} C + B$
 - **POS:** $(A + B) \cdot (A + \bar{B} + \bar{C}) \cdot C$
- These “mixed” forms are **neither SOP nor POS**
 - $(A B + C) (A + C)$
 - $A B \bar{C} + A C (A + B)$

Standard Sum-of-Products (SOP)

- A sum of minterms form for n variables can be written down directly from a truth table.
 - Implementation of this form is a two-level network of gates such that:
 - The first level consists of n -input AND gates, and
 - The second level is a single OR gate (with fewer than 2^n inputs).
- This form often can be simplified so that the corresponding circuit is simpler.

Standard Sum-of-Products (SOP)

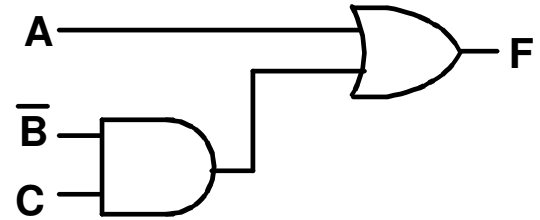
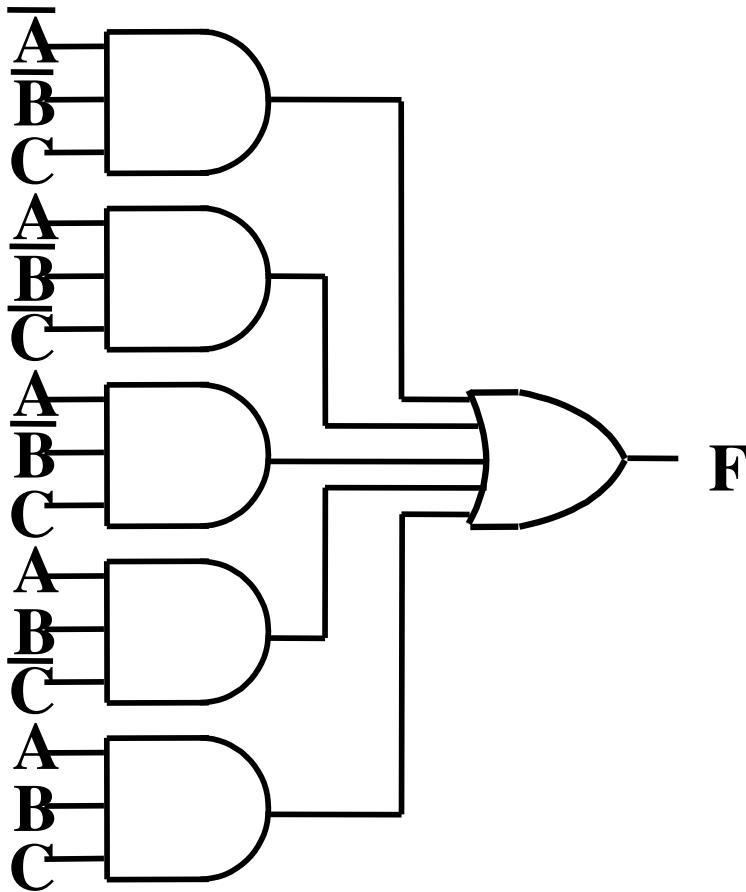
- **A Simplification Example:**
- **$F(A, B, C) = \Sigma m(1, 4, 5, 6, 7)$**
- **Writing the minterm expression:**
 $F = \overline{A} \overline{B} C + A \overline{B} \overline{C} + A \overline{B} C + ABC\overline{C} + ABC$
- **Simplifying:**

$$F = A + \overline{B}C$$

- **Simplified F contains 3 literals compared to 15 in minterm F**

AND/OR Two-level Implementation of SOP Expression

- The two implementations for F are shown below – it is quite apparent which is simpler!



- **The previous examples show that:**
 - **Canonical Forms (Sum-of-minterms, Product-of-Maxterms), or other standard forms (SOP, POS) differ in complexity**
 - **Boolean algebra can be used to manipulate equations into simpler forms.**
 - **Simpler equations lead to simpler two-level implementations**
- **Questions:**
 - **How can we attain a “simplest” expression?**
 - **Is there only one minimum cost circuit?**
 - **The next part will deal with these issues.**